

Interpretable Deep Probabilistic Model for HRR Radar Signal and its Application to Target Recognition

Leiyao Liao, Lan Du , Senior Member, IEEE, and Jian Chen 

Abstract—With the advent of neural network, unprecedented advancements have been achieved in many tasks, including radar signal processing. However, a key disadvantage of the current methods is the black-box structure, which makes it hard to interpret or assess the hidden representations of data. In this paper, by combining the physical generative mechanism of the high range resolution (HRR) radar signal with neural network, we develop an interpretable deep probabilistic model to learn the latent features from HRR radar signals that can characterize the physical structure of targets. In detail, based on the radar target's scattering center model which describes the HRR radar signal as the summation of echoes from the scattering centers, a deep probabilistic model is constructed to depict the generative process from the scattering centers to observations, where the latent features comprise the locations and amplitudes of scattering centers. Considering that the locations of scattering centers are nearly invariable and the amplitudes of them are fluctuant for the signals within a small angular range, our model defines the shared locations and independent amplitudes of scattering centers for the signals in a small angular range, aiming to improve the robustness of interpretable feature extraction model. In addition, our model performs probabilistic inference on the posterior distribution of latent features with high preciseness. With the proposed model, we further design a recognition scheme based on the minimum reconstruction error criterion. Experiments on the measured HRR radar dataset validate the effectiveness of our model on learning interpretable features and superior recognition performance.

Index Terms—Radar automatic target recognition (RATR), interpretability, neural network, variational inference, minimum reconstruction error.

I. INTRODUCTION

WITH the development of radar high-resolution technology, the high-dimensional signal received from the high range resolution (HRR) radar can be easily collected and processed. Moreover, the complex frequency response from target, i.e., the HRR radar signal, embodies the target structure and

electromagnetic characteristics, such as target size, scatterer distribution, and so on. Therefore, the HRR radar target recognition has elicited wide attention in the field of radar automatic target recognition (RATR), which belongs to the high-dimensional sensing technique [32]–[34] of radar signal processing.

A. Motivation of Interpretable Deep Probabilistic Model

Since original HRR radar signals are high-dimensional and may often contain some redundancy information, some feature extraction methods are developed to improve the processing efficiency and target recognition performance. Most of recent HRR radar target recognition methods adopt the amplitude of complex time domain echo from targets as the input signal, which is called the high-resolution range profile (HRRP). In [3], Du *et al.* adopt the principal component analysis (PCA) to convert HRRPs into the feature subspace and then represent them with several linear uncorrelated principal components. To precisely depict the HRRP, Feng *et al.* [4] learn the sparse overcomplete representations with the k -singularly valuable decomposition (K-SVD). Moreover, in [5], the factor analysis (FA) is utilized to model HRRPs and obtain several low-dimensional latent factors, which performs well on target recognition. With recent explosive growth of artificial intelligence, some deep learning (DL)-based methods have shown their advances in HRRP target recognition filed. In [6], Feng *et al.* develop a stacked corrective auto-encoder (SCAE) to learn the hierarchical features of HRRPs by taking advantage of the nonlinear neural network. In addition, Ying *et al.* [37] build a Robust Variational Auto-Encoder to extract the robust features for HRRPs by reconstructing the average profile in each HRRP frame. Du *et al.* [7] construct a probabilistic generative model named factorized discriminative conditional variational auto-encoder, which incorporates the labels to improve the discriminability of features. Moreover, Zhang *et al.* [38] propose a novel method to structure the discriminant feature and jointly learn the max-margin classifier a unified Bayesian, and the method has shown the effectiveness on HRRP target recognition. In [8], a Target-Aware Recurrent Attentional Network (TARAN) is designed to analyze the sequential dependence of the range cells in the HRRP, which gains promising recognition performance. In addition, Pan *et al.* [39] build an advanced framework that contains the nested CNN, stacked bidirectional recurrent neural network (bi-RNN), dynamic adjustment layer, squeeze and excitation (SE) block,

Manuscript received October 21, 2021; revised February 1, 2022; accepted March 1, 2022. Date of publication March 17, 2022; date of current version July 11, 2022. This work was supported in part by the National Science Foundation of China under Grant U21B2039 and Grant 61771362, and in part by 111 Project. The guest editor coordinating the review of this manuscript and approving it for publication was Dr. Xin Yuan. (Corresponding author: Lan Du.)

The authors are with the National Lab of Radar Signal Processing, Xidian University, Xi'an 710071, China (e-mail: liaoleiyao_mail@163.com; dulan@mail.xidian.edu.cn; qtehcjjan@163.com).

Digital Object Identifier 10.1109/JSTSP.2022.3160241

and multilayer attention mechanism, which devote to extract the representative features. These DL-based methods can effectively explore the data features that are beneficial to target recognition.

Despite the powerful performance gains, the above data-driven methods are of black-box structure and cannot be directly explained, which means these methods hide certain decision-making risks in real-world applications. Recently, some interpretable networks have been developed in various fields. In [29], Zhang *et al.* add a loss to the filters of the high layers in CNN to constraint the features maps being learned to represent specific parts of the targets, which can disentangle the latent features of the CNN, and experiments show the effective target detection results. Sabour *et al.* [30] develop the capsule network for image classification, in which the messages from the capsules in lower layers can be sent to the capsules in higher layers to encode specific semantic features. Moreover, Zhang *et al.* [31] build a deep interpretable network for the compressive image sensing by incorporating the AMP algorithm [20] into the network, aiming to unfold the latent representations learned by the network with specific meaning. Additionally, some interpretable networks have been proposed for RATR, and the main way of them is introducing attention modules to enhance the interpretability of network. Chen *et al.* [12] build a model by adapting the dual-channel mechanism of spatial attention and channel attention to realize target area extraction for SAR images. In [13], Li *et al.* equip the spatial pixel-level attention mechanism to neural network, which can guide the network to highlight the target region of interest and eliminate the background clutter in SAR images. Although the interpretable network with attention module for RATR gives explanations for the effect of neural network, they neglect the physical characteristics of radar data, such as statistical and semantic knowledge, which may restrict their performance.

To deeply excavate the essential characteristics of targets from observed data, in this paper, based on the physical model of HRR radar targets that the radar signals are approximated as a sum of responses from individual scattering centers, we build an interpretable deep probabilistic model to access the hidden representations of signals. The HRR radar signals are complex echoes from targets, thus we build our model with complex neural network. In our model, the physical generative process of HRR radar signals from the latent space to data space is precisely depicted, and the latent representations are learned to cover the locations and amplitudes of scattering centers of radar targets. According to the physical model, the locations of scattering centers for the signals in a small angular range hardly change, while the amplitudes of them are very sensitive to the variation of target-aspect angles. To precisely depict the physical characteristics of HRR radar signals, our model adopts the way of learning the shared locations and independent amplitudes of scattering centers for the signals in a small angular range, which improves the robustness of our model on scattering centers extraction. Moreover, the learning of latent features impacts the representative ability of a model. Specifically, our deep probabilistic model infers the posterior

distribution of latent features to obtain the features with high preciseness and further improves the model performance. In addition, we derive a complex back-propagation algorithm [15], [24] to effectively optimize the whole network. Furthermore, our model can be applied to the target recognition, and a recognition scheme is designed based on the minimum reconstruction error criterion.

B. Contributions

The principal contributions of this paper are listed as follows:

- We focus on the construction of an interpretable deep probabilistic model for the complex HRR radar signals by incorporating radar target's scattering center model into neural network to explore the latent features with specific physical meanings. Different from most of existing real-valued networks, our model is a complex network for the complex HRR radar signals. By modeling complex HRR radar signals, our model learns the latent features that represent the locations and amplitudes of scattering centers. Moreover, for the signals within a small angular range, the locations of scattering centers are shared and their amplitudes are learned independently in our model, in order to improve the preciseness of data description and robustness of features extraction.
- Our model is a deep probabilistic network, which can infer the posterior distribution of latent features to attain high accuracy of features. By the means of effective approximate inference on the latent variables, our model can automatically determine the number of the dominant scattering centers, which enhances the accuracy of scattering centers features extraction and further improves the representative ability of our model.
- Based on the minimum reconstruction error criterion, we design a recognition scheme with our model, which gains the superior recognition performance. By extracting the interpretable latent features that reflect the target physical structure information, our model performs well on target recognition and has good generalization ability even with the incomplete training set.

To assist the readers understand the definitions of notations easily, Table I lists all the notations in this paper.

The remainder of the research report here is organized as follows. Section II shows the physical forming process of HRR radar signals. Then, Section III shows a detailed introduction to the proposed model, and Section IV presents the recognition scheme of our model. Section V displays some experiments on the measured HRR radar dataset to evaluate the effectiveness of the proposed model. Section VI summarizes the conclusions of this paper.

II. PHYSICAL FORMING MECHANISM OF HRR RADAR SIGNAL

This section presents a detailed chain of obtaining the HRR radar signal. In general, the linear frequency modulated (LFM) signal [9], [15] is adopted as the transmitted signal $x_t(t)$ and is

TABLE I
LIST OF NOTATIONS IN THIS PAPER

Notation	Meaning	Notation	Meaning
$x_i(t)$	The transmitted signal	r_k	The locations of the k th range cells
$s(t)$	The signal envelope of $x_i(t)$	φ	The parameter of the inference network
Δf	The frequency interval between neighboring frequency components	$\mathbf{y}_n^{(d,m)}$	The m th HRR radar signal in the m th aspect-frame from the target d
T_p	The pulse width of a rectangular pulse waveform	$\mathbf{w}_n^{(d,m)}$	The scattering coefficient of the HRR radar signal $\mathbf{y}_n^{(d,m)}$
$\hat{x}_i(t)$	The received echo in the i th range cell	$\boldsymbol{\eta}_n^{(d,m)}$	The noise components in $\mathbf{y}_n^{(d,m)}$
ΔR	The range resolution	Φ	The Fourier basis
c	The propagation velocity of radar signal	$\sigma(k)$	The coefficient of k th scattering center
B	The bandwidth	D	The number of targets
M_d	The number of aspect-frames of d th target	$\hat{\mathbf{w}}_n^{(d,m)}$	The coefficient of scattering centers for $\mathbf{y}_n^{(d,m)}$
R_{ik}	The radial distance between radar and k th scattering center in i th range cell	$\mathbf{s}^{(d,m)}$	The location of scattering centers for signals in the m th aspect-frame of the target d
$\mathbf{z}_n$	The latent feature of $\mathbf{x}_n$	θ	The parameter of generative network
$y(l)$	The l th frequency response	$\boldsymbol{\mu}_n^{(d,m)}$	The mean of the variables $\hat{\mathbf{w}}_n^{(d,m)}$
L	The number of frequency components	$(\boldsymbol{\sigma}_n^{(d,m)})^2$	The variance of the variables $\hat{\mathbf{w}}_n^{(d,m)}$
μ	The FM coefficient	$\boldsymbol{\rho}^{(d,m)}$	The mean of $\mathbf{s}^{(d,m)}$
K	The number of range cells	$\mathbf{W}$	The weight in the inference network
N	The number of samples in an aspect-frame	$\mathbf{b}$	The bias in the inference network
P	The dimension of HRR radar signal $\mathbf{y}_n^{(d,m)}$	$\hat{\mathbf{y}}_n^{(d,m)}$	The generative data for the original data $\mathbf{y}_n^{(d,m)}$

expressed as

$$\begin{aligned} x_i(t) &= s(t) e^{j2\pi f_c t} \\ &= \text{rect}\left(\frac{t}{T_p}\right) \exp(j\pi\mu t^2) \exp(j2\pi f_c t) \end{aligned} \quad (1)$$

where $s(t) = \text{rect}(t/T_p) \exp(j\pi\mu t^2)$ represents the signal envelope, $j = \sqrt{-1}$ is the imaginary unit, $\text{rect}(t/T_p)$ represents a rectangular pulse waveform with pulse width T_p , f_c denotes the radar center frequency, μ denotes the LFM coefficient, and the bandwidth is $B = \mu T_p$.

As discussed in [1], [2], [9], [15], the response of a target can be regarded as the coherent summation of the responses from individual target scattering centers along radar line-of-sight. For a wideband radar system, the received signals can be divided into many range cells along the range dimension by the range resolution $\Delta R = c/2B$ with c denoting the propagation velocity of radar signal. Thus the radar signatures from the scattering centers within a range cell will be coherently summed into a single signature in that range cell. After the processing of

dechirping technique [27], [28], the received echo in the i th range cell ($i = 1, 2, \dots, M$) in baseband is written as

$$\begin{aligned} \hat{x}_i(t) &= \sum_{k=1}^L \sigma(k) s\left(t - \frac{2R_{ik}}{c}\right) \exp\left(-j2\pi f_c \frac{2R_{ik}}{c}\right) \\ &= \sum_{k=1}^L \sigma(k) \text{rect}\left(\frac{t - \frac{2R_{ik}}{c}}{T_p}\right) \\ &\quad \times \exp\left(-j2\pi f_c \frac{2R_{ik}}{c} + j\pi\mu\left(t - \frac{2R_{ik}}{c}\right)^2\right) \end{aligned} \quad (2)$$

where L represents the number of target scattering centers in i th range cell, $\sigma(k)$ represents the scattering coefficient of k th scattering center, R_{ik} represents the radial distance between radar and k th scattering center in i th range cell.

Moreover, by applying matched filtering technique to the signal in (2), the frequency response can be expanded as

$$\begin{aligned} y(f) &= F(\hat{x}_i(t)) \cdot s^*(f) \\ &= \sum_{k=1}^L \sigma(k) s(f) \exp(-j2\pi(f + f_c) \frac{2R_{ik}}{c}) s^*(f) \\ &= \sum_{k=1}^L \sigma(k) P(f) \exp(-j2\pi(f + f_c) \frac{2R_{ik}}{c}) \end{aligned} \quad (3)$$

where $F(\cdot)$ represents the Fourier transform operation, $s(f)$ denotes the Fourier transform of $s(t)$, and $s^*(f)$ represents the conjugate operation of $s(f)$, $P(f) = s(f)s^*(f)$ denotes the power spectrum of $s(t)$. For the LFM signal, we have $P(f) \approx \text{rect}(f/B)$, and (3) is further expressed as

$$\begin{aligned} y(f) &= \sum_{k=1}^L \sigma(k) \exp\left(-j2\pi f_c \frac{2R_{ik}}{c}\right) \exp\left(-j2\pi f \frac{2R_{ik}}{c}\right) \\ &= \sum_{k=1}^L w(k) \exp\left(-j2\pi f \frac{2R_{ik}}{c}\right) \end{aligned} \quad (4)$$

where $w(k) = \sigma(k) \exp(-j2\pi f_c \frac{2R_{ik}}{c})$. By discretization to (4), we can obtain the discrete frequency response

$$y(l) = \sum_{k=1}^L w(k) \exp\left(-j2\pi \frac{2R_{ik}}{c} (l-1) \Delta f\right) \quad (5)$$

where $y(l)$ denotes the l th frequency response, and $l \in \{1, 2, \dots, L\}$ with L representing the number of frequency components, Δf represents the frequency interval between the neighboring frequency components. Let's suppose that R_{ik} is an integer multiple of ΔR , and we can rewrite (5) as

$$y(l) = \sum_{k=1}^K w(k) \exp\left(-j2\pi \frac{2r_k}{c} (l-1) \Delta f\right) \quad (6)$$

where $w(k)$ is the scattering coefficient in the k th range cell, and $\{r_k\}_{k=1}^K$ represent the locations of the K range cells with $r_k = k \cdot \Delta R$. Furthermore, if we consider the noise in practical applications, the signal in (6) can be represented as

$$y(l) = \sum_{k=1}^K w(k) \exp\left(-j2\pi \frac{2r_k}{c} (l-1) \Delta f\right) + \eta(l) \quad (7)$$

with $\{\eta(l)\}_{l=1}^L$ denoting the noise components. Equation (7) shows the concrete expression of the HRR radar signal. Let

$$\phi(r_k) = [1, \exp(-j\frac{4\pi}{c}r_k\Delta f), \dots, \exp(-j\frac{4\pi}{c}r_k(L-1)\Delta f)]^T$$

$$\Phi = [\phi(r_1), \phi(r_2), \dots, \phi(r_K)] \quad (8)$$

Thus, the signal model in (7) can be transformed into a vector-matrix form as

$$\mathbf{y}_n = \Phi \mathbf{w}_n + \boldsymbol{\eta}_n \quad (9)$$

where $\mathbf{y}_n = [y_n(1), \dots, y_n(L)]^T$ denotes the HRR radar signal, $\mathbf{w}_n = [w_n(1), \dots, w_n(K)]^T$ denotes the scattering coefficient and $\boldsymbol{\eta}_n = [\eta_n(1), \dots, \eta_n(K)]^T$ represents the noise vector; Φ is the Fourier basis, and if $K = L$, Φ is represented as a complete orthogonal basis matrix; if $K > L$, Φ is an over-complete basis matrix.

As discussed above, the HRR radar signals are approximated by the echoes from the dominant scattering centers which are assumed to be sparsely distributed in the target, and the signals from weak scattering centers are usually considered as the noise components [19]. In addition, the HRR radar signals have three sensitivity issues that are needed to be considered, including translation, amplitude-scale and target-aspect sensitivities [1], [7], [40], [41]. In practical application, the first two issues can be respectively handled by translation alignment [40] and amplitude-scale normalization [41]. In addition, the target-aspect sensitivity is a key issue of HRR radar signals, which means that the scatterer distribution of a target varies with the target aspect. To handle the target-aspect sensitivity, the received signals covering multiple target-aspect angles are divided into many subsets, and the signals in a subset without scatterers' motion through range cells (MTRC) [2]–[5] are from a target-aspect sector [1], [7].

III. THE PROPOSED MODEL

A subset of the signals from a target-aspect sector without the scatterers' MTRC is called as an aspect-frame [9], [10]. Thus the signal model of the n th sample in the m th aspect-frame from the target d is expressed as

$$\mathbf{y}_n^{(d,m)} = \Phi \mathbf{w}_n^{(d,m)} + \boldsymbol{\eta}_n^{(d,m)} \quad (10)$$

where $n \in \{1, 2, \dots, N\}$ with N denoting the number of samples in an aspect-frame, and $m \in \{1, 2, \dots, M_d\}$ with M_d denoting the number of aspect-frames of d target, and $d \in \{1, 2, \dots, D\}$ with D denoting the number of targets.

To access the sparse distributed property of dominant scattering centers, the scattering coefficient can be decomposed as $\mathbf{w}_n^{(d,m)} = \hat{\mathbf{w}}_n^{(d,m)} \circ \mathbf{s}_n^{(d,m)}$, where $\circ$ denotes the point-wise (Hadamard) vector product, $\hat{\mathbf{w}}_n^{(d,m)}$ denotes the coefficient of scattering centers in all range cells, and $\mathbf{s}_n^{(d,m)}$ represents a sparse binary vector. The value of the k th dimension of the variable $\mathbf{s}_n^{(d,m)}$ reflects whether or not the dominant scattering centers locate in the k th range cell. In addition, according to the analysis in [1], [2], the locations of scattering centers for

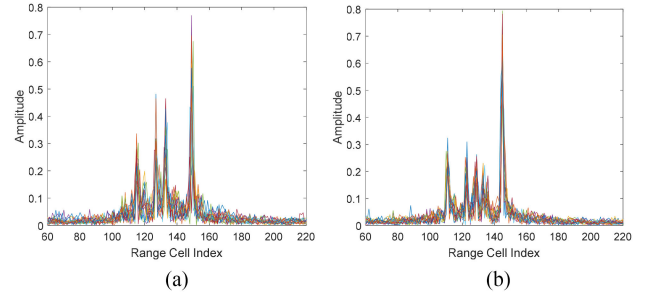


Fig. 1. The HRRPs from Yark-42. (a) The signals in the target-aspect range $2.16^\circ \sim 2.61^\circ$. (b) The signals in the target-aspect range $4.08^\circ \sim 4.41^\circ$.

the signals within an aspect-frame is a stabilize term and does hardly change with target-aspect angles; while the amplitudes are very sensitive to the variation of target-aspect angles, and thus the amplitudes are fluctuant even with small change of target-aspect angles. Fig. 1 shows two examples of some HRRPs in the small range of target-aspect angles (about 0.4°) from Yark-42, which illustrates that the locations of scattering centers are similar, while the amplitudes of them are fluctuant. Thus, we define the shared locations and independent amplitudes of scattering centers for the signals in an aspect-frame, and express the scattering coefficient as $\mathbf{w}_n^{(d,m)} = \hat{\mathbf{w}}_n^{(d,m)} \circ \mathbf{s}^{(d,m)}$. Then the (10) can be rewritten as

$$\mathbf{y}_n^{(d,m)} = \Phi \left(\hat{\mathbf{w}}_n^{(d,m)} \circ \mathbf{s}^{(d,m)} \right) + \boldsymbol{\eta}_n^{(d,m)} \quad (11)$$

where $\mathbf{s}^{(d,m)}$ reflects the shared location of scattering centers for signals in the m th aspect-frame of the target d . Usually, (11) is referred to the radar target's scattering center model.

Combining the above physical model with neural network, we build an interpretable deep probabilistic model for HRR radar signals in an aspect-frame, which consists of generative network and inference network modules. In the following, a detailed introduction to the two modules will be presented, and the objective function for model training also will be shown.

A. The Generative Network

Due to the remarkable gains of neural network in various fields, we incorporate the above radar target's scattering center model into the neural network and define a generative network. The latent variables $\{\hat{\mathbf{w}}_n^{(d,m)}, \mathbf{s}^{(d,m)}\}$ are fed to generative network to generate the data $\hat{\mathbf{y}}_n^{(d,m)}$. To be more specific, the latent variables $\{\hat{\mathbf{w}}_n^{(d,m)}, \mathbf{s}^{(d,m)}\}$ are fed to the latent layer, and at the bottom layer the data $\hat{\mathbf{y}}_n^{(d,m)}$ is generated:

$$\text{Latentlayer} : \tilde{\mathbf{h}}_n^{(d,m)} = \Phi \left(\hat{\mathbf{w}}_n^{(d,m)} \circ \mathbf{s}^{(d,m)} \right) \quad (12)$$

$$\text{DataGeneration} : \mathbf{y}_n^{(d,m)} \sim p_\theta(\tilde{\mathbf{h}}_n^{(d,m)}, \boldsymbol{\eta}_n^{(d,m)}) \quad (13)$$

As shown in (12), the generative network is implemented by the multi-layer perceptron (MLP) structure with a single linear layer, in which the network parameter θ is fixed as the Fourier basis Φ . The architecture of the generative network is shown in Fig. 2. If the Fourier basis Φ is complete orthogonal basis matrix,

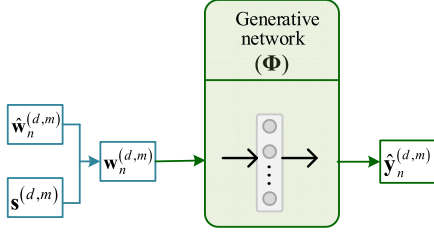


Fig. 2. Architecture of the generative network in our model.

the dimension of the latent feature is the same as that of the input data; if the Fourier basis Φ is over-complete, the generative network can explore the super-resolution representation of data. In (13), for the sake of simplicity, we adopt complex Gaussian distribution for $p_\theta(\cdot)$.

In the developed generative network, the unknown variables include $\{\hat{\mathbf{w}}_n^{(d,m)}, \mathbf{s}^{(d,m)}, \boldsymbol{\eta}_n^{(d,m)}\}$. For the generative network, the prior distributions of latent variables are set as follows. In the light of the central limit theorem, the summation of some independent variables can be approximated to follow the Gaussian distribution. Since the complex echo from a range cell is the coherent summations of returns from scattering centers in the range cell, the real and imaginary parts of the complex echo are approximately regarded as Gaussian distribution. Thus, the coefficient $\hat{\mathbf{w}}_n^{(d,m)}$ is formulated as a K -dimensional complex Gaussian distribution:

$$\hat{\mathbf{w}}_n^{(d,m)} \sim \text{CNorm}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \quad (14)$$

where $\text{CNorm}(\cdot)$ denotes the complex Gaussian distribution, and (14) is further expanded as

$$p_0(\hat{\mathbf{w}}_n^{(d,m)}) = \frac{1}{(\pi)^K |\boldsymbol{\Sigma}_0|} \exp \left\{ -\left(\hat{\mathbf{w}}_n^{(d,m)} - \boldsymbol{\mu}_0 \right)^H \boldsymbol{\Sigma}_0^{-1} \left(\hat{\mathbf{w}}_n^{(d,m)} - \boldsymbol{\mu}_0 \right) \right\} \quad (15)$$

where the mean and covariance are respectively set as $\boldsymbol{\mu}_0 = 0$, and $\boldsymbol{\Sigma}_0 = \mathbf{I}_K$ for ease of inference. Then to seek for the sparsity on the vector $\mathbf{s}^{(d,m)}$, the j th dimension of it is modeled as

$$s_0^{(d,m)}(j) \sim \text{Bernoulli}(\rho_0) \quad (16)$$

where $j \in \{1, \dots, K\}$ with K representing the dimension of the sparse vector, and ρ_0 denotes the prior parameter of Bernoulli distribution that determines the sparsity degree. Moreover, as depicted in some literatures [10], [22], the noises in the radar signals of aircraft targets are assumed as the Gaussian white noises. Thus, the noise is assumed to be drawn *i.i.d* from a complex Gaussian distribution with zero-mean and unknown variance that can be estimated by some pre-processing methods [23]. Thus the noise component is modeled as

$$\boldsymbol{\eta}_n^{(d,m)} \sim \text{CNorm}(\mathbf{0}, \sigma^2 \mathbf{I}) \quad (17)$$

with σ^2 denoting the variance.

Under the generative network defined in Eqs. (12)–(17), the joint log-likelihood of the observed data and the latent variables

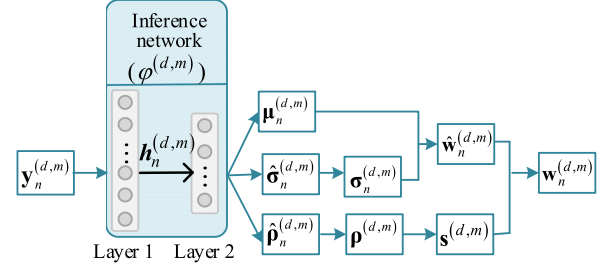


Fig. 3. Architecture of the inference network in our model.

is expressed as

$$\begin{aligned} \log p \left(\mathbf{y}_n^{(d,m)}, \hat{\mathbf{w}}_n^{(d,m)}, \mathbf{s}^{(d,m)} | \Phi, \mathbf{w}_n^{(d,m)}, \sigma^2, \boldsymbol{\mu}_n^{(d,m)}, \right. \\ \left. \boldsymbol{\Sigma}_n^{(d,m)}, \boldsymbol{\rho}^{(d,m)} \right) = \log \left[p \left(\mathbf{y}_n^{(d,m)} | \Phi, \mathbf{w}_n^{(d,m)}, \sigma^2 \right) \right. \\ \left. \times p \left(\hat{\mathbf{w}}_n^{(d,m)} | \boldsymbol{\mu}_n^{(d,m)}, \boldsymbol{\Sigma}_n^{(d,m)} \right) \right] + \log p \left(\mathbf{s}^{(d,m)} | \boldsymbol{\rho}^{(d,m)} \right) \end{aligned} \quad (18)$$

where $\{\boldsymbol{\mu}_n^{(d,m)}, \boldsymbol{\Sigma}_n^{(d,m)}\}$ and $\boldsymbol{\rho}^{(d,m)}$ denote the posterior parameters of the variables $\hat{\mathbf{w}}_n^{(d,m)}$ and $\mathbf{s}^{(d,m)}$, respectively. The proposed generative network is constructed with the physical forming mechanism of HRR radar signals, and the latent states represent the amplitudes and locations of scattering centers.

B. The Inference Network

The goal of the generative network is to maximize marginal log-likelihood of observations to infer the posterior parameters of the latent variables. However, it is usually intractable to compute the marginal distribution directly, since the solving of the true posteriors requires an integral operation that is hard to access. Here we adopt an alternative way by using a deep probabilistic inference network to infer the approximated posteriors, which projects the data into the latent space to obtain the latent representations of observed data.

Firstly, for the latent variables $\{\hat{\mathbf{w}}_n^{(d,m)}, \mathbf{s}^{(d,m)}\}$, we introduce the fully factorized approximate posterior distributions as $q(\hat{\mathbf{w}}_n^{(d,m)} | \mathbf{y}_n^{(d,m)})$ and $q(\mathbf{s}^{(d,m)} | \mathbf{y}_n^{(d,m)})$. For the coefficient $\hat{\mathbf{w}}_n^{(d,m)}$, it is modeled as a complex Gaussian distribution, and its approximate posterior is expressed as

$$q \left(\hat{\mathbf{w}}_n^{(d,m)} | \mathbf{y}_n^{(d,m)} \right) = \text{CNorm} \left(\boldsymbol{\mu}_n^{(d,m)}, \text{diag} \left(\left(\boldsymbol{\sigma}_n^{(d,m)} \right)^2 \right) \right) \quad (19)$$

Analogously, the posterior distribution of sparse vector $\mathbf{s}^{(d,m)}$ is

$$q \left(\mathbf{s}^{(d,m)} | \mathbf{y}_n^{(d,m)} \right) = \text{Bernoulli} \left(\boldsymbol{\rho}^{(d,m)} \right) \quad (20)$$

Furthermore, the inference network with multiply nonlinear layers is employed to infer the whole posterior parameters $\{\boldsymbol{\mu}_n^{(d,m)}, (\boldsymbol{\sigma}_n^{(d,m)})^2, \boldsymbol{\rho}^{(d,m)}\}$.

To illustrate the inference network in its simplest form, here we consider the neural network with two layers, and Fig. 3 presents the architecture of the inference network.

Different from the generative network being top-down, the inference network is bottom-up, starting at the layer 1 with the input data $\mathbf{y}_n^{(d,m)}$: (21)–(29) shown at the bottom of this page, where $\tanh(\cdot)$ denotes the hyperbolic tangent function. The network parameters are $\varphi = \{\mathbf{W}_1^{(d,m)}, \mathbf{W}_2^{(d,m)}, \mathbf{W}_3^{(d,m)}, \mathbf{W}_4^{(d,m)}, \mathbf{b}_1^{(d,m)}, \mathbf{b}_2^{(d,m)}, \mathbf{b}_3^{(d,m)}, \mathbf{b}_4^{(d,m)}\}$ with $\{\mathbf{W}_1^{(d,m)}, \mathbf{W}_2^{(d,m)}, \mathbf{W}_3^{(d,m)}, \mathbf{W}_4^{(d,m)}\}$ denoting the weight matrices and $\{\mathbf{b}_1^{(d,m)}, \mathbf{b}_2^{(d,m)}, \mathbf{b}_3^{(d,m)}, \mathbf{b}_4^{(d,m)}\}$ the biases. Due to the complex-valued nature of HRR radar signals, the inference network is complex, and the complex feed-forward propagation [24], [25] is derived, as listed in (21)–(24). Since the real and imaginary parts of the output can be negative, the activate function is chosen as the hyperbolic tangent function here, which is adjustable according to the data type. In addition, for the complex Gaussian distribution of $\hat{\mathbf{w}}_n^{(d,m)}$, we define $|\hat{\sigma}_n^{(d,m)}|$ as its standard deviation $\sigma_n^{(d,m)}$ via (25). For (28), coefficient $\hat{\mathbf{w}}_n^{(d,m)}$ is sampled as $\hat{\mathbf{w}}_n^{(d,m)} = \mu_n^{(d,m)} + \sigma_n^{(d,m)} \circ \varepsilon$ with $\varepsilon \sim \text{CNorm}(\mathbf{0}, \mathbf{I})$ via the reparameterization technique [14], [15]. Similarly, the parameter of Bernoulli distribution is real-valued, and thus we define the $|\hat{\rho}_n^{(d,m)}|$ via (26). Then, in (27), the average of the N variables $\{\hat{\rho}_n^{(d,m)}\}_{n=1}^N$ is operated to obtain the parameter of Bernoulli distribution, representing the posterior values of the shared location of scattering centers for the samples in m th aspect-frame. Moreover, we define that the sparse location vector $\mathbf{s}^{(d,m)}$ is sampled from $\rho^{(d,m)}$ via (29), and the value in each dimension means the probability of containing center scattering in each range cell. More specifically, to simplify the model training, we approximate $\mathbf{s}^{(d,m)}$ with its

Algorithm 1: The Training Stage of the Proposed Model.

- 1: Set the architecture of our model, including the dimension of latent variables and the number of layers;
 - 2: Initialize the inference network parameter φ and fix the generative network parameter as Φ ;
 - 3: **while** not converged **do**
 - 4: Randomly sample a mini-batch $\mathbf{Y}_1^{(d,m)} = \{\mathbf{y}_n^{(d,m)}\}_{n=1}^{N_1}$ from the data in m th aspect-frame from d th target;
 - 5: Sample random noise from the standard complex Gaussian distribution for reparameterization;
 - 6: Get the posterior distributions of latent states $\hat{\mathbf{w}}_n^{(d,m)}$ and $\mathbf{s}^{(d,m)}$ from $\text{CNorm}(\mu_n^{(d,m)}, \text{diag}((\sigma_n^{(d,m)})^2))$ and $\text{Bernoulli}(\rho^{(d,m)})$ via (28) and (29), respectively; and then generate observation $\mathbf{y}_n^{(d,m)}$ from $\text{CNorm}(\hat{\mathbf{h}}_n^{(d,m)}, \eta_n^{(d,m)})$ via (13);
 - 7: Update φ via SGD on the objective function, i.e., (30).
 - 8: **end while**
-

mean value $\rho^{(d,m)}$, which allows back-propagation through the variable $\mathbf{s}^{(d,m)}$. Due to the sparsity of the variable $\mathbf{s}^{(d,m)}$, most of the values in $\rho^{(d,m)}$ are close to 0. Moreover, the variance of the Bernoulli variable $\mathbf{s}^{(d,m)}$ that is calculated as $\rho^{(d,m)}(1 - \rho^{(d,m)})$ is close to 0. Therefore, the way of approximating $\mathbf{s}^{(d,m)}$ with its mean $\rho^{(d,m)}$ is reasonable.

$$\begin{aligned} \text{Output in layer 1: } \mathbf{h}_n^{(d,m)} &= \tanh \left(\Re \left(\mathbf{W}_1^{(d,m)} \mathbf{y}_n^{(d,m)} + \mathbf{b}_1^{(d,m)} \right) \right) \\ &\quad + j \tanh \left(\Im \left(\mathbf{W}_1^{(d,m)} \mathbf{y}_n^{(d,m)} + \mathbf{b}_1^{(d,m)} \right) \right) \end{aligned} \quad (21)$$

$$\text{Output1 in layer 2 : } \mu_n^{(d,m)} = \mathbf{W}_2^{(d,m)} \mathbf{h}_n^{(d,m)} + \mathbf{b}_2^{(d,m)} \quad (22)$$

$$\begin{aligned} \text{Output2 in Layer 2: } \hat{\sigma}_n^{(d,m)} &= \tanh \left(\Re \left(\mathbf{W}_3^{(d,m)} \mathbf{h}_n^{(d,m)} + \mathbf{b}_3^{(d,m)} \right) \right) \\ &\quad + j \tanh \left(\Im \left(\mathbf{W}_3^{(d,m)} \mathbf{h}_n^{(d,m)} + \mathbf{b}_3^{(d,m)} \right) \right) \end{aligned} \quad (23)$$

$$\begin{aligned} \text{Output3 in layer 2: } \hat{\rho}_n^{(d,m)} &= \tanh \left(\Re \left(\mathbf{W}_4^{(d,m)} \mathbf{h}_n^{(d,m)} + \mathbf{b}_4^{(d,m)} \right) \right) \\ &\quad + j \tanh \left(\Im \left(\mathbf{W}_4^{(d,m)} \mathbf{h}_n^{(d,m)} + \mathbf{b}_4^{(d,m)} \right) \right) \end{aligned} \quad (24)$$

$$\text{Modular operation: } \sigma_n^{(d,m)} = |\hat{\sigma}_n^{(d,m)}| \quad (25)$$

$$\text{Modular operation: } \bar{\rho}_n^{(d,m)} = |\hat{\rho}_n^{(d,m)}| \quad (26)$$

$$\text{Average operation : } \rho^{(d,m)} = \frac{1}{N} \sum_{n=1}^N \bar{\rho}_n^{(d,m)} \quad (27)$$

$$\text{Code Generation : } \hat{\mathbf{w}}_n^{(d,m)} \sim \text{CNorm} \left(\mu_n^{(d,m)}, \text{diag} \left((\sigma_n^{(d,m)})^2 \right) \right) \quad (28)$$

$$\text{Code Generation: } \mathbf{s}^{(d,m)} \sim \text{Bernoulli} \left(\rho^{(d,m)} \right) \quad (29)$$

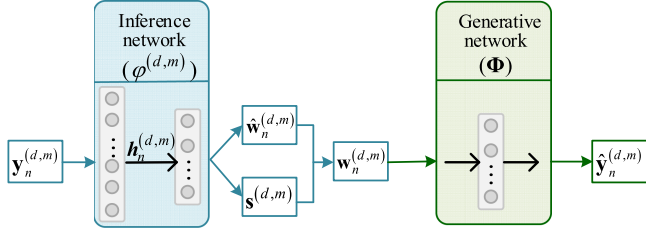


Fig. 4. The overall architecture of our model. The observed data $\mathbf{y}_n^{(d,m)}$ is input to the inference to obtain $\hat{\mathbf{w}}_n^{(d,m)}$ and $\mathbf{s}^{(d,m)}$, and then gain the $\mathbf{w}_n^{(d,m)}$, which is then fed to the generative network to obtain the data $\hat{\mathbf{y}}_n^{(d,m)}$. The detailed process of obtaining the $\hat{\mathbf{w}}_n^{(d,m)}$ and $\mathbf{s}^{(d,m)}$ is eliminated here.

C. Objective Function and Learning Process

Fig. 4 shows the overall architecture of our proposed model. In the following, we present a method for learning the inference network jointly with the generative network.

Firstly, we derive the total objective function of our model as

$$\begin{aligned}
 LB = & \mathbb{E}_{q(\hat{\mathbf{w}}_n^{(d,m)}|\mathbf{y}_n^{(d,m)}), q(\mathbf{s}^{(d,m)}|\mathbf{y}_n^{(d,m)})} \left[\log p(\mathbf{y}_n^{(d,m)}|\mathbf{w}_n^{(d,m)}) \right] \\
 & - \text{KL} \left[q(\hat{\mathbf{w}}_n^{(d,m)}|\mathbf{y}_n^{(d,m)}) \parallel p(\hat{\mathbf{w}}_n^{(d,m)}) \right] \\
 & - \text{KL} \left[q(\mathbf{s}^{(d,m)}|\mathbf{y}_n^{(d,m)}) \parallel p(\mathbf{s}^{(d,m)}) \right]
 \end{aligned} \quad (30)$$

where $\text{KL}(q(\cdot)||p(\cdot)) = \int q(\cdot) \ln \frac{q(\cdot)}{p(\cdot)}$ is the Kullback-Leibler (KL) divergence that measures the deviation degree of the approximate $q(\cdot)$ from the prior $p(\cdot)$ and is non-negative with its value equal to zero only when $q(\cdot) = p(\cdot)$. The first term on the right hand side of (30) denotes the expectation of the log likelihood $p(\mathbf{y}_n^{(d,m)}|\mathbf{w}_n^{(d,m)})$ and the other terms on the right hand side act as the regularization for Gaussian and Bernoulli variables respectively. The complete derivation and detailed solution of (30) can be found in the **Appendix**. Due to the complex-valued nature of the HRR radar signal, our model is built with complex neural network. Thus, for the objective function in (30), we derive a complex back-propagation algorithm [15], [24] based on the stochastic gradient descent (SGD) to optimize the network parameters, and the total training process is outlined in Algorithm 1.

IV. THE IMPLEMENTATION OF TARGET RECOGNITION

For the application of target recognition, the training dataset contains the signals from D targets and is represented as $\{\mathbf{y}_n^{(d,m)}\}_{d=1, m=1, n=1}^{D, M_d, N}$. As described in Section III, by modeling the data from each aspect-frame via our model, there are $\sum_{d=1}^D M_d$ inference networks for the whole observations. Furthermore, with the proposed model, we develop a target recognition scheme based on the minimum reconstruction error criterion. Specifically, after the training process, the network parameters are fixed, and there are $\sum_{d=1}^D M_d$ trained location templates that can be represented as $\{\mathbf{s}^{(d,m)}\}_{d=1, m=1}^{D, M_d}$. With the trained model, the recognition process by frame templates

Algorithm 2: The Recognition Procedure of the Proposed Method.

- 1: Obtain the test data $\mathbf{y}^*$
- 2: **for** $d = 1 : D$
- for** $m = 1 : M_d$
- 3: Input the test data $\mathbf{y}^*$ into the inference network trained by the data from the m th aspect-frame of the target d , and obtain the latent state $\hat{\mathbf{w}}_*^{(d,m)}$ from $q(\hat{\mathbf{w}}_*^{(d,m)}|\mathbf{y}^*)$ via reparameterization trick;
- 4: Multiply the coefficient $\hat{\mathbf{w}}_*^{(d,m)}$ with the locations templates $\mathbf{s}^{(d,m)}$ to obtain the scattering coefficient $\mathbf{w}_*^{(d,m)}$
- 5: Fed the scattering coefficient $\mathbf{w}_*^{(d,m)}$ into the generative model and gain the corresponding reconstructed sample $\hat{\mathbf{y}}_*^{(d,m)}$;
- 6: Calculate the reconstruction error between reconstructed sample $\hat{\mathbf{y}}_*^{(d,m)}$ and the original test sample $\mathbf{y}^*$ via $\|\mathbf{y}^* - \hat{\mathbf{y}}_*^{(d,m)}\|_2^2$;
- 7: **end for**
- 8: **end for**
- 9: Obtain class index d^* via (31), and then classify the test sample $\mathbf{y}^*$ to the d^* th target class.

matching will be presented. To give an intuitive interpretation, Fig. 5(a) shows the recognition architecture for the training phase and test phase (recognition procedure). In addition, to effectively reduce the network parameters, layer sharing in the inference networks is further adopted in the recognition method, and the architecture of inference networks with several shared layers is shown in Fig. 5(b). Therefore, the recognition model can be jointly optimized with all observations.

Given a test sample $\mathbf{y}^*$, the coefficient $\hat{\mathbf{w}}_*^{(d,m)}$ is obtained from $q(\hat{\mathbf{w}}_*^{(d,m)}|\mathbf{y}^*)$ by the feed-forward propagation via the inference networks. With the location template $\mathbf{s}^{(d,m)}$, we gain the scattering coefficient $\mathbf{w}_*^{(d,m)} = \hat{\mathbf{w}}_*^{(d,m)} \circ \mathbf{s}^{(d,m)}$. After that, we can obtain the reconstructed signal $\hat{\mathbf{y}}_*^{(d,m)}$ via the generative network. With $\sum_{d=1}^D M_d$ trained location templates, we obtain a set of reconstructions $\{\hat{\mathbf{y}}_*^{(1,1)}, \dots, \hat{\mathbf{y}}_*^{(d,m)}, \dots, \hat{\mathbf{y}}_*^{(D,M)}\}$. Then the reconstruction errors between the test sample $\mathbf{y}^*$ and reconstruction $\{\hat{\mathbf{y}}_*^{(1,1)}, \dots, \hat{\mathbf{y}}_*^{(d,m)}, \dots, \hat{\mathbf{y}}_*^{(D,M)}\}$ can be calculated. According to the minimum reconstruction error criterion, the test sample $\mathbf{y}^*$ is classified to the d^* th class with

$$d^* = \arg \min_{d \in \{1, \dots, D\}, m \in \{1, \dots, M_d\}} \left\| \mathbf{y}^* - \hat{\mathbf{y}}_*^{(d,m)} \right\|_2^2 \quad (31)$$

where $\|\cdot\|_2$ denotes the L2-norm. The total recognition process of our method is outlined in Algorithm 2.

V. EXPERIMENTS

A. Data Description and Experimental Setup

In this section, we adopt a widely used measured dataset [3]–[10] to verify the effectiveness of our model. The signals in the

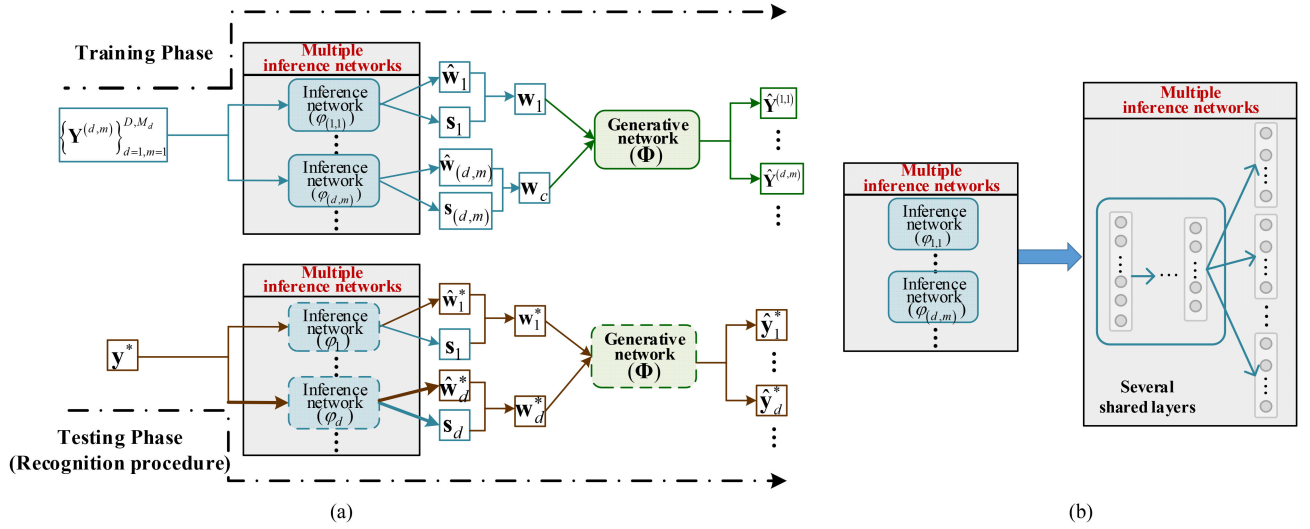


Fig. 5. Architecture of our recognition method. (a) The architecture of model training phase and test phase (recognition procedure). The solid boxes of the networks indicate that the parameters are learned, while the dashed boxes mean that the network parameters are fixed. In the training phase, the blue arrows indicate the feed-forward propagation of training data in inference networks, and the green arrows indicate the feed-forward propagation of training data in generative network. In the test phase, the blue arrows indicate the fixed location templates for test data, while the brown arrows indicate the feed-forward propagation of test data. Moreover, the location templates come from the training phase, and the coefficients are obtained via feed-forward propagation of test data. (b) The architecture of the inference networks with layer sharing.

TABLE II
PARAMETERS OF THREE AIRPLANES AND RADAR IN THE EXPERIMENT

radar parameters	center frequency		5520 MHz
	bandwidth		400 MHz
planes	length (m)	width (m)	height (m)
Yark-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

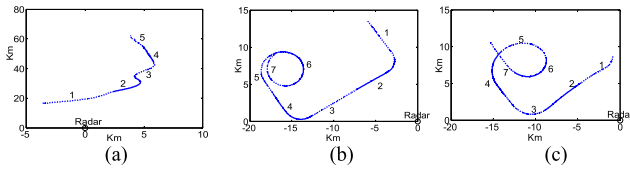


Fig. 6. The track charts of radar targets. (a) Yark-42. (b) Cessna Citation S/II. (c) An-26.

dataset are received from three airplane targets, including Yark-42, Cessna Citation and An-26, via the cooperative experiments under high signal-to-noise ratio ($SNR \approx 40dB$). Table II lists the measurements of radar and the parameters of the targets. In Fig. 6, we plot the two-dimensional track charts of the three targets with each target track being split into several segments. The 2th and 5th segments from Yark-42, the 6th and 7th segments from Cessna Citation S/II, and the 5th and 6th segments from An-26 are utilized as training data, and other data segments are taken as test data [15], [21]–[22]. Moreover, the signals from each target are divided into some subsets, and each subset is defined as an aspect-frame. In the dataset, Yark-42, Cessna Citation S/II

and An-26 targets respectively have 35, 50 and 50 aspect-frames, i.e., $D = 3$, $M_1 = 35$, $M_2 = M_3 = 50$, and each aspect-frame has 1024 samples, i.e., $N = 1024$, and the dimension of each sample is 256, i.e., $L = 256$.

The befitting architecture of inference network is chosen as 256–300–400–256 when $K = L$ and 256–300–400–512 when $K = 2L$ for the dataset. For the recognition model, it contains 135 ($M_1 + M_2 + M_3 = 135$) inference networks for the dataset, and the network parameters of the first two shallow layers are shared in the inference networks. In the training process, we initialize the real and imaginary parts of network weights by sampling from standard Gaussian and the bias terms as 0; we set the batchsize as 32, and set the learning rate of the SGD algorithm as 10^{-3} . Unless specified otherwise, the above setting is adopted across all the experiments.

B. Feature Extraction Performance

Most of neural network-based methods are of black-box structure and learn the abstract features, but our model is an interpretable network that can learn the features with physical meanings, and the features cover the amplitudes and locations of the scattering centers. In this section, we conduct some experiments to analyze the interpretability of latent features.

Firstly, we show the effect of our model on learning the location variable $s^{(d,m)}$. If there is a dominant scattering center in the k th range cell, the posterior value of $s_k^{(d,m)}$ should be one, otherwise it is zero. The results in the first row of Fig. 7 are the locations inferred by our models with $K = L$ for samples in an aspect-frame from Yark-42 and the second row shows the corresponding samples. As shown in the top two rows of Fig. 7, the inferred locations by our model can

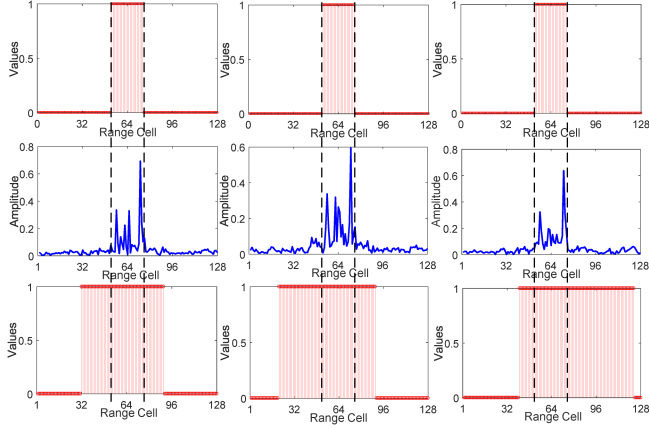


Fig. 7. The HRRPs and inferred locations. The first row shows the locations inferred by our model with $K = L$ for three samples in an aspect-frame from Yark-42, the second row shows the corresponding HRRPs samples in the aspect-frame, and the third row shows the inferred locations by the model without location sharing.

precisely locate the dominant scattering centers for the samples. Since our model infers the posterior of the location variable $s^{(d,m)}$, the number of scattering centers can be automatically determined and then the locations can be inferred accurately. For the model without scattering center's location sharing, the results are presented in the third row of Fig. 7, which shows that the inferred locations vary with samples and are not accurate to locate scattering centers. Thus, Fig. 7 illustrates that our model can learn the locations for the data in an aspect-frame with high accuracy.

Then we analyze the feature extraction performance of our model, comparing to some related methods, including the model without location sharing, orthogonal matching pursuit (OMP) algorithm [9], relaxation (RELAX) [42], and Bayesian Compressive Sensing (BCS) [43]. Fig. 8 shows the learned features of two test samples and their reconstructions. The results in first column of Fig. 8 are obtained by our model, in which the extracted latent features represent the locations and amplitudes of scattering centers that can depict the signals precisely, and the corresponding reconstructions are very close to the original signals. For the results in the second column of Fig. 8 gained by the model without scattering center's location sharing, the learned features perform bad to describe the signals and the reconstructions diverge greatly from the original signals. The results in the first and second columns validate the effectiveness of our model with the location sharing. The third to fifth columns in Fig. 8 display the results achieved by the OMP, RELAX and BCS, and we can see that some scattering centers are neglected, and the reconstructed samples are not the same as the original signals. The performance of OMP, RELAX and BCS are very sensitive to the termination threshold of iteration that is hard to obtain the accurate values, and thus their performance are restricted. Being a probabilistic network that infers the posterior distributions of the scattering coefficients, our model automatically determines the number of scattering centers. Thus our model can extract the scattering features precisely and obtain good data reconstruction results.

Furthermore, to compare the reconstruction performance of our models with some methods, i.e., the model without location sharing, OMP [9], RELAX [42], and BCS [43], Fig. 9 presents the reconstructions of a test sample. In Fig. 9, the method with $K = 2L$ can better reconstruct some details of signals than the corresponding method with $K = L$, and our method performs better than other methods. Moreover, we give the quantitative results with root mean square error (RMSE), defined as $\sqrt{\|\hat{\mathbf{x}}^* - \mathbf{x}^*\|_2^2 / \|\mathbf{x}^*\|_2^2}$ with $\hat{\mathbf{x}}^*$ denoting the original test signal and $\hat{\mathbf{x}}$ the reconstructed signal. The results of test samples from different targets by different methods are listed in Table III, which shows that our model has the minimum RMSEs for all targets, and the RMSEs of test data achieved by the model with $K = 2L$ are much smaller than those obtained by the model with $K = L$. Thus, the results in Fig. 9 and Table III both verify the effectiveness of our model on data reconstruction.

C. Recognition Performance

In this section, we conduct some experiments to evaluate the recognition performance of our method, in comparison to some related methods, including the traditional approaches: the linear discriminant analysis (LDA) [26], k-singularly valuable decomposition (K-SVD) [4], linear SVM (LSVM) [17], adaptive Gaussian classifier (AGC) [18], principal component analysis (PCA) [3], Hausdorff distance-based OMP (HD-OMP) [9], Bayesian-model [10], and the deep learning approaches: neural network (NN), stacked denoising auto-encoder (SDAE) [33], stacked corrective auto-encoders (SCAE) [6], Target-Aware Recurrent Attentional Network (TARAN) [8], tensor RNN with Dirichlet process mixture (DPmTRNN) [36].

1) *Experiments With Complete Data:* In the experiments with complete data, our model is learned with the training dataset covering the complete aspect-frames. After model training, we perform test procedure and present the test recognition results of our method and several related methods in Fig. 10, which lists the confusion matrices and average recognition rates. Each row in each figure of Fig. 10 is the predicted ratio of test samples in a specific class. As we can see from Fig. 10, the recognition rate of our model with $K = 2L$ for the target An-26 is lower than some other methods, but our method has much higher recognition rates for Cessna and Yark-42, and thus the average recognition performance of our model with $K = 2L$ is much better than other methods. Since more precise description for data is explored by the model with $K = 2L$, more useful information can be utilized to improve target recognition performance.

Moreover, we compare our method with some related recognition methods and the results are summarized in Table IV. As presented in Table IV, the deep models can achieve better recognition performance than the shallow models, since the neural network can learn more representative features from the observed data. Among deep models, our model outperforms other models, especially the model with $K = 2L$ gains superior result. By combining the physical forming mechanism of HRR radar signals with neural network, our model can learn the interpretable features to excavate the essential target structure

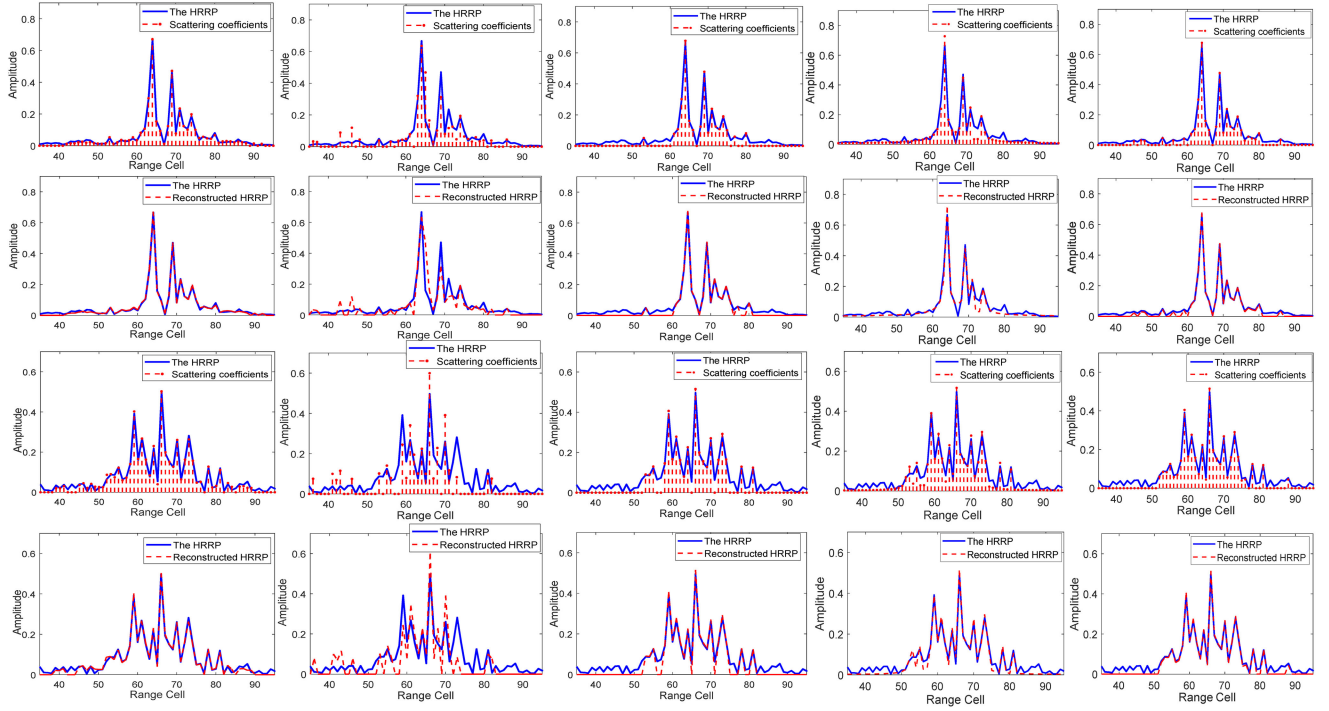


Fig. 8. The extracted latent features and reconstructions of two test samples by some methods. The results in the five columns from left to right are obtained by our model, the model without location sharing, the OMP, RELAX and BCS when $K = L$, respectively. The top two rows are the results of a test sample from Cessna Citation S/II, and the bottom two rows are the results of a test sample from Yark-42.

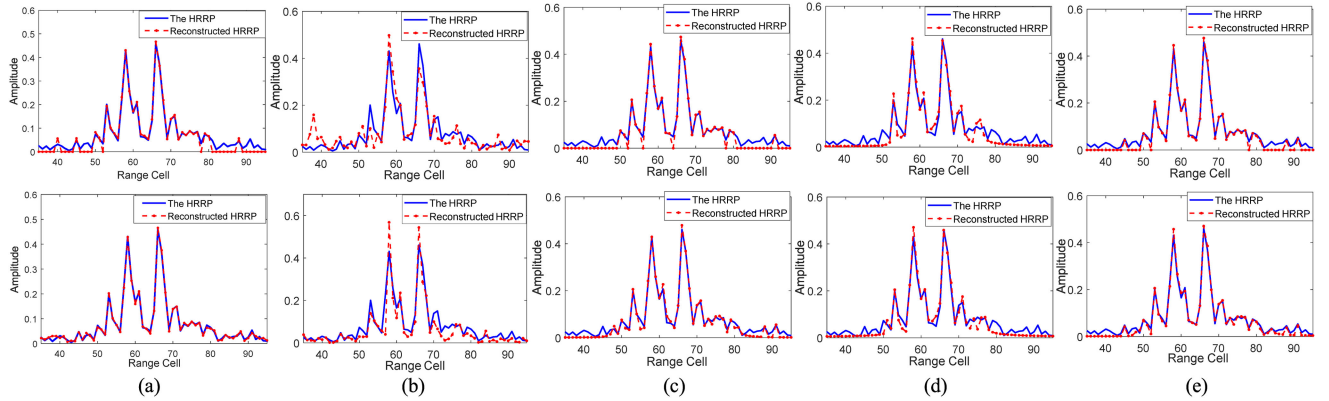


Fig. 9. The reconstructions of one test sample by different methods. The results in (a)–(e) are obtained by our model, the model without location sharing, the OMP, RELAX and BCS, respectively. The first row of (a)–(e) Shows the results obtained by the methods with $K = L$, and the second row of (a)–(e) Shows the results obtained by the methods with $K = 2L$.

TABLE III
THE RMSES OF TEST SAMPLES WITH DIFFERENT METHODS

Methods \ Targets	Our model		The model without location sharing		OMP		RELAX		BCS	
	$K = L$	$K = 2L$	$K = L$	$K = 2L$	$K = L$	$K = 2L$	$K = L$	$K = 2L$	$K = L$	$K = 2L$
An-26	0.0478	0.0384	0.0978	0.0755	0.0764	0.0595	0.0746	0.0706	0.0439	0.0327
Cessna Citation S/II	0.0623	0.0516	0.1525	0.1364	0.0876	0.0730	0.1051	0.0850	0.0685	0.0562
Yark-42	0.1029	0.0851	0.1929	0.1684	0.1237	0.1024	0.1758	0.1631	0.1159	0.1135
Average RMSEs for all targets	0.0710	0.0584	0.1477	0.1268	0.0959	0.0783	0.1185	0.1062	0.0761	0.0604

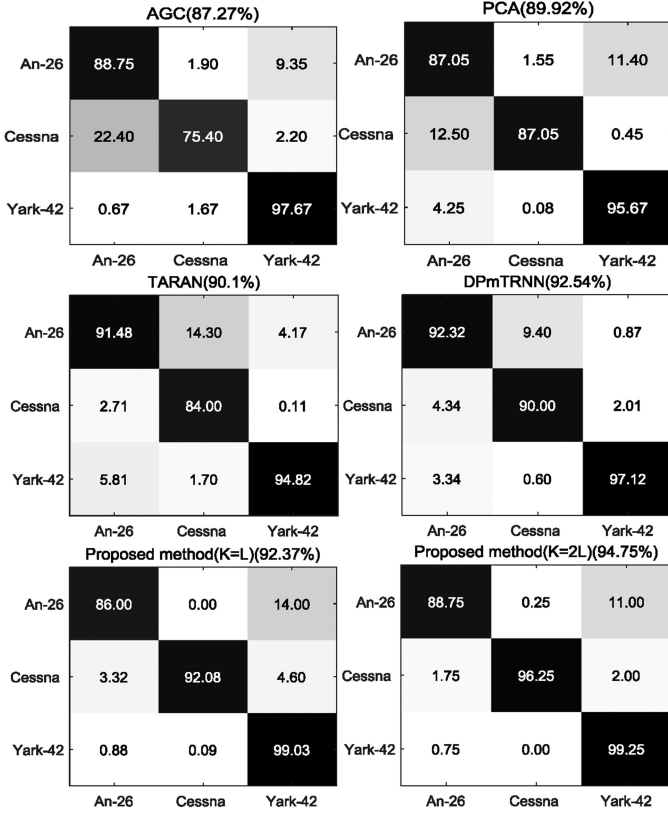


Fig. 10. The confusion matrix comparison of different methods.

TABLE IV
RECOGNITION RESULTS OF OUR METHOD AND SOME RELATED METHODS

Methods	Accuracy (%)
LDA[26]	82.16
K-SVD[4]	85.32
LSVM[32]	85.63
AGC[18]	87.08
PCA[3]	89.92
HD-OMP ($K = L$)[9]	87.08
HD-OMP ($K = 2L$)[9]	87.18
Bayesian-model ($K = L$)[10]	85.72
Bayesian-model ($K = 2L$)[10]	90.01
NN	90.20
SDAE[33]	90.04
SCAE[6]	92.03
TARAN [8]	90.10
DPmTRNN [36]	92.54
Our method ($K = L$)	92.37
Our method ($K = 2L$)	94.75

characteristics that are useful for target recognition. Moreover, our model can adopt the over-complete basis matrix to attain more precise representations of signals, achieving much better recognition performance.

In addition, to investigate the effect of scattering center's location sharing in a frame and layer sharing in the inference networks on the recognition performance of our method, we

TABLE V
THE RECOGNITION RATE (%) RESULTS OF ABLATION STUDY

Scattering center's location sharing in a frame	×	×	✓	✓
Layer sharing in the inference networks	×	✓	×	✓
$K = L$	82.71	84.36	89.63	92.37
$K = 2L$	85.22	86.53	91.94	94.75

conduct the ablation study and Table V lists the results. As shown in Table V, scattering center's location sharing in a frame is very effective to improve the recognition performance of our method, since the latent features can be extracted to depict the observed data with high preciseness. In addition, the results in Table V show that layer sharing in the inference networks can effectively improve the recognition performance of our method. Layer sharing in the inference networks can reduce model parameters and avoid over-fitting, which is beneficial to improve the representative ability of model. Therefore, scattering center's location sharing in a frame and layer sharing in the inference networks both devote to the superior recognition results of our methods.

2) *Results With Incomplete Data:* In aforementioned experiments, the target-aspect angles in the training dataset are complete. However, in some severe non-cooperative circumstance, some target-aspects are often deficient, and the available training dataset is very limited. In these cases, the generalization ability is an important criterion to evaluate the practicability of a model.

To analyze the generalization ability of our model, Fig. 11 shows the results of the inferred scattering coefficients for the samples in different aspect-frames from Cessna Citation S/II. Fig. 11(a)–(d) present four samples that are respectively in 4th, 5th, 7th and 9th aspect-frames. When the interval between two aspect-frames is smaller, the samples in the two frames are more similar. Fig. 11(e)–(h) display the inferred scattering coefficients by the model learned with the data from 4th, 5th, 7th and 9th aspect-frames, which shows that the inferred locations can accurately locate the dominant scattering centers and the amplitudes can precisely depict the original samples. Moreover, Fig. 11(i)–(k) show the inferred scattering coefficients by the model learned with the data from 4th aspect-frame. In detail, in Fig. 11(i), the inferred locations are accurate, but the amplitudes are a litter different from that in Fig. 11(f), such as the scattering center *A* in Fig. 11(i) and the scattering center *B* in Fig. 11(f). In Fig. 11(j), some of the inferred scattering centers are inaccurate, such as the scattering centers *D* and *F*, their amplitudes are different from those of the scattering centers *C* and *E* in Fig. 11(g). Moreover, it is obvious that the results in Fig. 10(h) are worse, such as the three scattering centers, *H*, *J* and *L*. Thus, we can see from Fig. 11(j) and (k), as the increase of the interval between frames, the performance of scattering center extraction for the samples in 5th, 7th and 9th frames gradually decreases. In addition, Fig. 12 presents the reconstruction results of three samples in 5th, 7th and 9th aspect-frames, and Fig. 12(a)–(c)

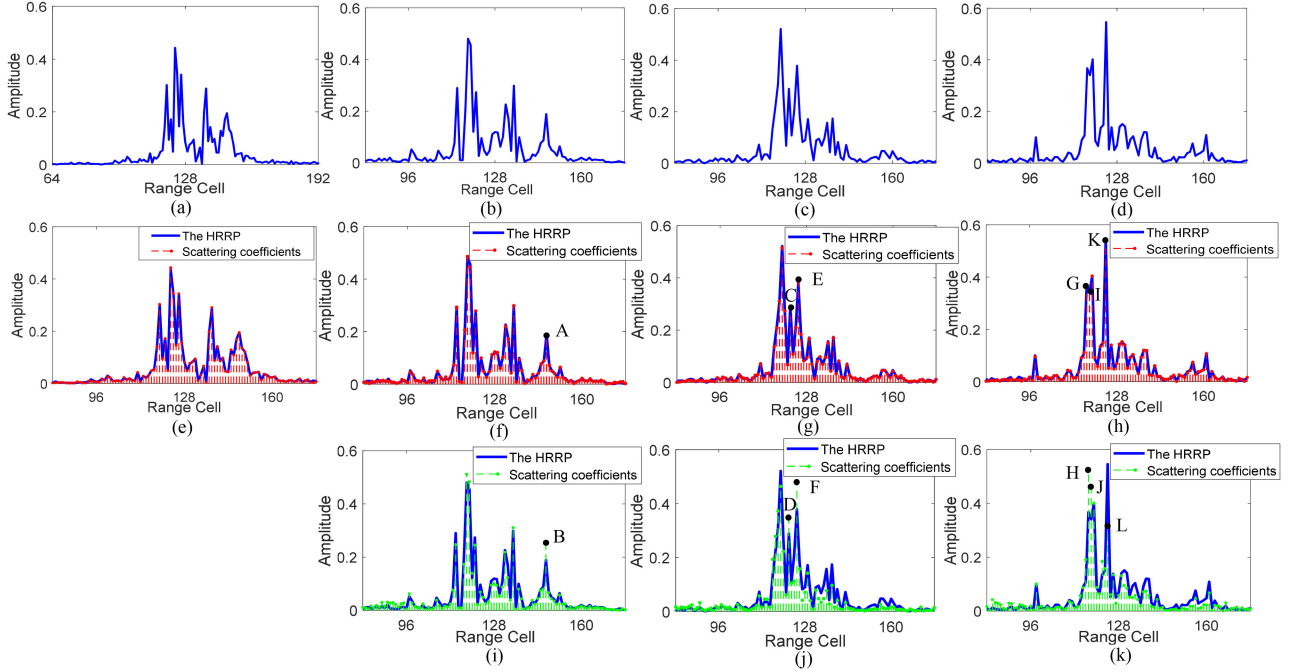


Fig. 11. The HRRP samples and the inferred scattering coefficients. (a)–(d) Show four samples in 4th, 5th, 7th and 9th aspect-frames from Cessna Citation S/II, respectively. (e)–(h) Show the inferred scattering coefficients by the model learned with the data from 4th, 5th, 7th and 9th aspect-frames, respectively. (i)–(k) Show the inferred scattering coefficients by the model learned with the data from 4th aspect-frame.

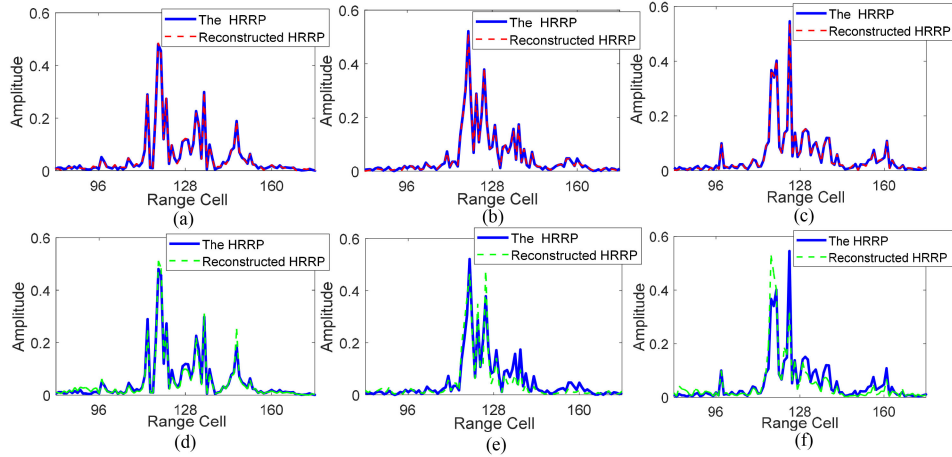


Fig. 12. The reconstructions for the three samples in Fig. 10(b)–(d), which are in 5th, 7th and 9th aspect-frames from Cessna Citation S/II. (a)–(c) are obtained by the model learned with the data from 5th, 7th and 9th aspect-frames, respectively. (d)–(f) are obtained by the model learned with the data from 4th aspect-frame.

are respectively obtained by the model learned with the data from 5th, 7th and 9th aspect-frames, while Fig. 12(d)–(f) are achieved by the model learned with the data from 4th aspect-frame. Fig. 12(a)–(c) show that the reconstructed samples are very similar to the original samples. In Fig. 12(d)–(f), as the interval between frames increases, the reconstruction effect gradually deteriorates with more peak values in the reconstructed samples being inaccurate. Thus, for neighboring several aspect-frames, the performance of our model on scattering center extraction and data reconstruction gradually decreases as the increase of the interval between frames, but our model can still attain

acceptable results, illustrating the good generalization ability of our model to describe unseen data in neighboring several aspect-frames.

To further verify the generalization ability of our model on recognition performance, we train our model with incomplete datasets, and the interval strides of selecting aspect-frames are 1, 2, 3, 4 and 5, defined as $\Delta = 1$, $\Delta = 2$, $\Delta = 3$, $\Delta = 4$ and $\Delta = 5$ for brevity. Fig. 13 presents the test recognition rates of our method and some related methods. As the increase of the interval strides of selecting aspect-frames, the recognition rates of all methods decrease. Moreover, the performance

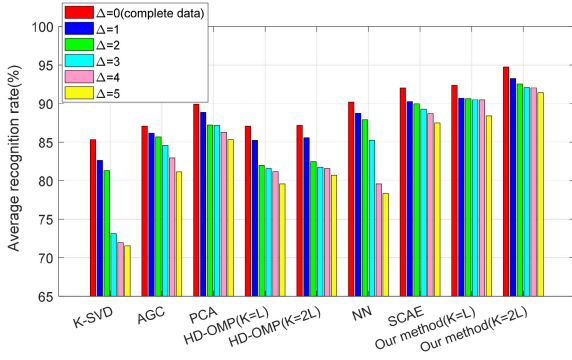


Fig. 13. Comparison of the generalization ability (recognition results) for our method and some related methods.

of the shallow methods, including K-SVD, AGC, PCA and HD-OMP, decreases quickly as the increase of interval strides, especially the K-SVD and HD-OMP. Since AGC and PCA are the Bayesian probabilistic models, they are more robust to the change of aspect-frames in training dataset. Moreover, for the deep methods, NN decreases rapidly as the increase of the interval strides, while SCAE and our model can still maintain the promising performance. SCAE and our model construct the models for the data from each aspect-frame to provide more precise description for the observations, which further leads to the robustness of the models. In our model, we incorporate the radar target's scattering center model into the neural network to explore physical characteristics of HRR radar signals, and the learned scattering center model of an aspect-frame has good generalization ability for describing the data from neighboring several aspect-frames. Comparing to SCAE being of black-box structure, our model can attain better recognition performance when the training data contain limited aspect-frames, verifying the good generalization ability of our model on recognition.

D. Influence of Model Parameters

The above experiments of our model are performed under the befitting architecture. The two key points of the network architecture are the number of latent layers and the number of latent feature nodes. Thus it is important to analyze the impact of the two network architecture parameters on the recognition performance of our model.

For our proposed model, the number latent layers can be set according to the data type, here we choose several cases to analyze its impact on the. Moreover, the number of latent feature nodes K in our model represents the dimension of the latent features, here we consider two cases of the models with complete orthogonal basis matrix and the over-complete basis matrix, and thus the latent features nodes are set as $K = L$ and $K = 2L$. Experiments are conducted to show the impact of model parameters on the test recognition results.

In Fig. 14, we present the test results of our model under different architectures as a function of training iterations. For the parameter of the number of latent layers, it is well-known

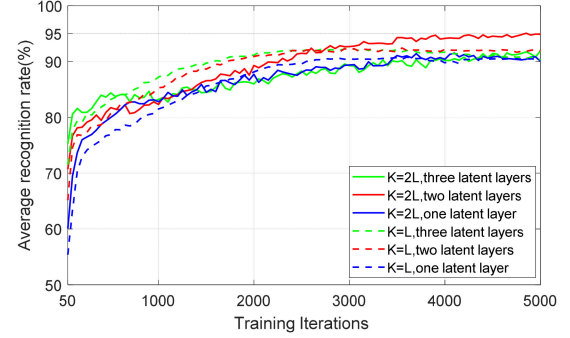


Fig. 14. The test recognition performance as a function of training iterations for the proposed model under different architectures.

TABLE VI
THE RECOGNITION RATES (%) OF OUR MODEL UNDER DIFFERENT ARCHITECTURES

Model structure	$K = L$			$K = 2L$		
	One latent layer	Two latent layers	Three latent layers	One latent layer	Two latent layers	Three latent layers
Average recognition rate (%)	90.34	92.37	90.43	90.67	94.75	91.33

that the more the number of latent layers is, the more representative latent features our model can learn, and the more training data the network requires to converge. In this sense, the number of training data available determines the depth of a network. As shown in Fig. 14, the more the number of latent layers is, the slower the model converges. If the number of latent layers is too large, there are a large number network parameters, and thus the model tends to be overfitting, then the recognition performance will decrease. A tradeoff between the recognition results and convergence speed should be considered. As presented in Fig. 14, with regards to our model, two latent layers guarantee the promising performance on the measured data. Moreover, the dimension of latent features impacts the representative ability of models. Generally, the more the latent feature nodes are, the better recognition performance our model achieves, but the slower convergence speed the model is. In Fig. 14, the model with $K = 2L$ converges slower but gains better results after convergence than that with the model $K = L$. In addition, Table VI lists the quantitative recognition rates of the models under different architectures after convergence, from which we can see the befitting architecture of our model is set as two latent layers and the latent feature nodes with $K = 2L$.

Furthermore, we analyze the model parameters, training and test time for different methods. In the second column of Table VII, we show the expressions of the total number of free parameters for each model, where P denotes the dimension of the input signal, and L_1, L_2, L_3 respectively denote the number of nodes of three latent layers (our model and SCAE both contain three latent layers). Since our model is constructed for the data in each aspect-frame, it contains more parameters than SCAE [6] that constructs a model for overall data. Analogously, PCA

TABLE VII
THE TOTAL NUMBERS OF FREE PARAMETERS, TRAINING AND TEST TIME FOR THE PROPOSED MODEL AND SOME RELATED MODELS

Method	The numbers of model parameters	Training an epoch (s)	Test a sample (s)
PCA	$\sum_{d=1}^D \sum_{m=1}^{M_d} \left(\frac{P^2 + P}{2} + P \right)$	8.8994	0.3647
FA	$\sum_{d=1}^D \sum_{m=1}^{M_d} (PV_{opt} + P + 1)$ V_{opt} denotes the optimal number of dictionary atoms for each frame in the FA model.	39.6328	0.3750
K-SVD	$\sum_{k=1}^K (k^2 P + kP + k^3 + k^2) + P$ K denotes the number of iterations in OMP.	3.6699	0.1236
SCAE	$2(PL_1 + L_1 L_2 + L_2 L_3)$	50.4542	0.0167
Our model ($K = L$)	$PL_1 + L_1 L_2 + \sum_{d=1}^D \sum_{m=1}^{M_d} (P(L_2 + P))$	123.8365	0.0094
Our model ($K = 2L$)	$PL_1 + L_1 L_2 + \sum_{d=1}^D \sum_{m=1}^{M_d} (2P(L_2 + P))$	131.3427	0.0124

[3] and FA [5] are also used to model the data from each aspect-frame, while K-SVD [4] learns the dictionary for overall data. Due to the sparsity of HRR radar signal, the number of iteration is usually much less than P . Thus, K-SVD has less parameters than PCA and FA. Moreover, the third column of Table VII lists the time of training an epoch for different methods, and the results show that our model takes much more time for training, especially the model with $K = 2L$.

In real applications, for an off-line learning model, the time cost in the test stage is significant to evaluate the practicability of the model. In the last column of Table II, we present the time cost for a test sample, which shows that the test time of our model is much less than PCA, FA, K-SVD and SCAE. For the statistical models including PCA and FA, they need to infer the parameter of a Bayesian classifier [3], [5], which is usually time-consuming. Moreover, K-SVD takes some time to extract the features of a test sample via the OMP. In the test phase, the SCAE needs to learn a classifier, such as the linear support vector machine (SVM) [35], while our model can obtain the recognition results directly by the means of feed-forward propagation, which has high time-efficiency.

VI. CONCLUSION

We design an interpretable model to explore the features with physical meanings by incorporating the physical model of the HRR radar signals into neural network, and the latent features cover the locations and amplitudes of scattering centers for signals. The proposed model learns the shared locations of scattering centers for the signals within a certain target-aspect range that leads to the robustness of our model on interpretable features extraction. Moreover, our model provides an effective inference way on the latent features by inferring the posterior distribution of features, devoting to the high accuracy of the extracted features. A recognition scheme is further presented for the developed model based on the minimum reconstruction error criteria. Experiments on the measured data verify the interpretability of the latent features and the effectiveness of our model on target recognition.

APPENDIX

We provide a derivation for variational lower bound of the marginal log-likelihood (30) in the main paper).

$$\begin{aligned}
& \log p(\mathbf{y}_n^{(d,m)}) \\
&= \log \int q(\mathbf{w}_n^{(d,m)} | \mathbf{y}_n^{(d,m)}) \frac{p(\mathbf{y}_n^{(d,m)} | \mathbf{w}_n^{(d,m)}) p(\mathbf{w}_n^{(d,m)})}{q(\mathbf{w}_n^{(d,m)} | \mathbf{y}_n^{(d,m)})} d\mathbf{w}_n^{(d,m)} \\
&\geq \int q(\mathbf{w}_n^{(d,m)} | \mathbf{y}_n^{(d,m)}) \log \frac{p(\mathbf{y}_n^{(d,m)} | \mathbf{w}_n^{(d,m)}) p(\mathbf{w}_n^{(d,m)})}{q(\mathbf{w}_n^{(d,m)} | \mathbf{y}_n^{(d,m)})} d\mathbf{w}_n^{(d,m)} \\
&\mathbf{w}_n^{(d,m)} = \hat{\mathbf{w}}_n^{(d,m)} \circ \mathbf{s}^{(d,m)} \quad E_{q(\hat{\mathbf{w}}_n^{(d,m)}, \mathbf{s}^{(d,m)} | \mathbf{y}_n^{(d,m)})} \\
&\left[\log p(\mathbf{y}_n^{(d,m)} | \mathbf{w}_n^{(d,m)}) \right] \\
&- KL \left(q(\hat{\mathbf{w}}_n^{(d,m)}, \mathbf{s}^{(d,m)} | \mathbf{y}_n^{(d,m)}) \parallel p(\hat{\mathbf{w}}_n^{(d,m)}, \mathbf{s}^{(d,m)}) \right) \triangleq LB
\end{aligned} \tag{A.1}$$

We assume that the hidden variables $\hat{\mathbf{w}}_n^{(d,m)}$ and $\mathbf{s}^{(d,m)}$ are independent, and then

$$\begin{aligned}
& q(\hat{\mathbf{w}}_n^{(d,m)}, \mathbf{s}^{(d,m)} | \mathbf{y}_n^{(d,m)}) \\
&= q(\hat{\mathbf{w}}_n^{(d,m)} | \mathbf{y}_n^{(d,m)}) q(\mathbf{s}^{(d,m)} | \mathbf{y}_n^{(d,m)})
\end{aligned} \tag{A.2}$$

Thus, the ELBO is

$$\begin{aligned}
LB &= E_{q(\hat{\mathbf{w}}_n^{(d,m)}, \mathbf{s}^{(d,m)} | \mathbf{y}_n^{(d,m)})} \left[\log p(\mathbf{y}_n^{(d,m)} | \mathbf{w}_n^{(d,m)}) \right] \\
&- KL \left[q(\hat{\mathbf{w}}_n^{(d,m)} | \mathbf{y}_n^{(d,m)}) \parallel p(\hat{\mathbf{w}}_n^{(d,m)}) \right] \\
&- KL \left[q(\mathbf{s}^{(d,m)} | \mathbf{y}_n^{(d,m)}) \parallel p(\mathbf{s}^{(d,m)}) \right]
\end{aligned} \tag{A.3}$$

Moreover, the two terms of the KL divergence in the (A.3) are expanded as

$$\begin{aligned}
& KL \left[q(\hat{\mathbf{w}}_n^{(d,m)} | \mathbf{y}_n^{(d,m)}) \parallel p(\hat{\mathbf{w}}_n^{(d,m)}) \right] \\
&= \int q(\hat{\mathbf{w}}_n^{(d,m)} | \mathbf{y}_n^{(d,m)}) \ln q(\hat{\mathbf{w}}_n^{(d,m)} | \mathbf{y}_n^{(d,m)}) d\hat{\mathbf{w}}_n^{(d,m)} \\
&- \int q(\hat{\mathbf{w}}_n^{(d,m)} | \mathbf{y}_n^{(d,m)}) \ln p(\hat{\mathbf{w}}_n^{(d,m)}) d\hat{\mathbf{w}}_n^{(d,m)}
\end{aligned} \tag{A.4}$$

with

$$\begin{aligned}
& \int q(\hat{\mathbf{w}}_n^{(d,m)} | \mathbf{y}_n^{(d,m)}) \ln q(\hat{\mathbf{w}}_n^{(d,m)} | \mathbf{y}_n^{(d,m)}) d\hat{\mathbf{w}}_n^{(d,m)} \\
&= \int q(\hat{\mathbf{w}}_n^{(d,m)} | \mathbf{y}_n^{(d,m)}) \ln CN(\hat{\mathbf{w}}_n^{(d,m)} | \boldsymbol{\mu}_n^{(d,m)}, \\
&\quad \text{diag}((\boldsymbol{\sigma}_n^{(d,m)})^2)) d\hat{\mathbf{w}}_n^{(d,m)} \\
&= -K \ln(\pi) - \sum_k \left(1 + \log(\sigma_n^{(d,m)}(k))^2\right) \quad (\text{A.5}) \\
&\int q(\hat{\mathbf{w}}_n^{(d,m)} | \mathbf{y}_n^{(d,m)}) \ln p(\hat{\mathbf{w}}_n^{(d,m)}) d\hat{\mathbf{w}}_n^{(d,m)} \\
&= \int q(\hat{\mathbf{w}}_n^{(d,m)} | \mathbf{y}_n^{(d,m)}) \ln CN(\hat{\mathbf{w}}_n^{(d,m)} | \mathbf{0}, \mathbf{I}) dz \\
&= -K \ln(\pi) - \sum_k \left((\mu_n^{(d,m)}(k))^2 + (\sigma_n^{(d,m)}(k))^2\right) \quad (\text{A.6})
\end{aligned}$$

Therefore,

$$\begin{aligned}
& \text{KL} \left[q(\hat{\mathbf{w}}_n^{(d,m)} | \mathbf{y}_n^{(d,m)}) \parallel p(\hat{\mathbf{w}}_n^{(d,m)}) \right] \\
&= - \sum_k \left(1 + \log(\sigma_n^{(d,m)}(k))^2 - (\mu_n^{(d,m)}(k))^2 \right. \\
&\quad \left. - (\sigma_n^{(d,m)}(k))^2 \right) \quad (\text{A.7})
\end{aligned}$$

Meanwhile, we can obtain

$$\begin{aligned}
& \text{KL} \left[q(\mathbf{s}^{(d,m)} | \mathbf{y}_n^{(d,m)}) \parallel p(\mathbf{s}^{(d,m)}) \right] \\
&= \sum_{k=1}^K \rho_0 \log \frac{\rho_0}{\rho_k^{(d,m)}} + (1 - \rho_0) \log \frac{1 - \rho_0}{1 - \rho_k^{(d,m)}} \quad (\text{A.8})
\end{aligned}$$

where ρ_0 denotes the parameter of Bernoulli prior distribution that determines the sparsity degree, and $\rho_k^{(d,m)}$ represents the average activation of k th hidden neuron. Referring to (A.4) in combination with (A.7) and (A.8), the eventual ELBO is rewritten as

$$\begin{aligned}
LB &= E_{q(\hat{\mathbf{w}}_n^{(d,m)}, \mathbf{s}^{(d,m)} | \mathbf{y}_n^{(d,m)})} \left[\log p(\mathbf{y}_n^{(d,m)} | \mathbf{w}_n^{(d,m)}) \right] \\
&+ \sum_{k=1}^K \left(1 + \log(\sigma_n^{(d,m)}(k))^2 - (\mu_n^{(d,m)}(k))^2 \right. \\
&\quad \left. - (\sigma_n^{(d,m)}(k))^2 \right) \\
&- \sum_{k=1}^K \rho_0 \log \frac{\rho_0}{\rho_k^{(d,m)}} + (1 + \rho_0) \log \frac{1 - \rho_0}{1 - \rho_k^{(d,m)}} \quad (\text{A.9})
\end{aligned}$$

REFERENCES

- [1] J. B. Keller, "Geometrical theory of diffraction," *J. Opt. Soc. Amer.*, vol. 52, no. 2, pp. 116–130, Feb. 1962.
- [2] A. De Freitas and J. P. de Villiers, "Multiple scatterer tracking in high range resolution radar," in *Proc. 15th IEEE Int. Conf. Inf. Fusion*, Jul. 2012, pp. 1683–1688.
- [3] L. Du, H. W. Liu, Z. Bao, and J. Zhang, "Radar automatic target recognition using complex high-resolution range profiles," *IET Radar Sonar Navigation*, vol. 1, no. 1, pp. 18–26, Feb. 2007.
- [4] B. Feng, L. Du, H. W. Liu, and F. Li, "Radar HRRP target recognition based on K-SVD algorithm," in *Proc. IEEE CIE Int. Conf. Radar*, Oct. 2012, pp. 642–645.
- [5] L. Du, H. Liu, P. Wang, B. Feng, M. Pan, and Z. Bao, "Noise robust radar HRRP target recognition based on multitask factor analysis with small training data size," *IEEE Trans. Signal Process.*, vol. 60, no. 7, pp. 3546–3559, Apr. 2012.
- [6] B. Feng, B. Chen, and H. Liu, "Radar HRRP target recognition with deep networks," *Pattern Recognit.*, vol. 61, pp. 379–393, Jan. 2017.
- [7] C. Du, B. Chen, B. Xu, D. Guo, and H. Liu, "Factorized discriminative conditional variational auto-encoder for radar HRRP target recognition," *Signal Process.*, vol. 158, pp. 176–189, May 2019.
- [8] B. Xu, B. Chen, J. Wan, H. Liu, and L. Jin, "Target-aware recurrent attentional network for radar HRRP target recognition," *Signal Process.*, vol. 155, pp. 268–280, Feb. 2019.
- [9] L. Du, H. He, L. Zhao, and P. H. Wang, "Noise robust radar HRRP target recognition based on scatterer matching algorithm," *IEEE Sensor J.*, vol. 16, no. 6, pp. 1743–1753, Nov. 2016.
- [10] L. Du, P. Wang, L. Zhang, H. He, and H. Liu, "Robust statistical recognition and reconstruction scheme based on hierarchical Bayesian learning of HRR radar target signal," *Expert Syst. Appl.*, vol. 42, no. 14, pp. 5860–5873, Aug. 2015.
- [11] Z. Huang, D. M., and Z. Pan, "Deep SAR-net: Learning objects from signals," *ISPRS J. Photogrammetry Remote Sens.*, vol. 161, pp. 179–193, Mar. 2020.
- [12] L. Chen, S. Tan, and Z. Pan, "A new framework for automatic airports extraction from SAR images using multi-level dual attention mechanism," *Remo. Sens.*, vol. 12, no. 3, pp. 560–584, Jan. 2020.
- [13] C. Li, L. Du, and S. Deng, "Point-wise discriminative auto-encoder with application on robust radar automatic target recognition," *Signal Process.*, vol. 169, Apr. 2020, Art. no. 107385.
- [14] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," in *Proc. Int. Conf. Learn. Represented*, 2014, pp. 1–14.
- [15] L. Liao, L. Du, and J. Chen, "Class factorized complex variational auto-encoder for HRR radar target recognition," *Signal Process.*, vol. 182, no. 1, May 2021, Art. no. 107932.
- [16] H. L. P. Vincent, I. Lajoie, Y. Bengio, and P. Manzagol, "Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion," *J. Mach. Learn. Res.*, vol. 11, no. 12, pp. 3371–3408, Dec. 2010.
- [17] C. Chang and C. Lin, "LIBSVM: A library for support vector machines," *ACM Trans. Intell. Syst. Technol.*, vol. 2, no. 3, pp. 1–27, May 2011.
- [18] J. Hu, L. Du, H. He, and P. H. Wang, "Noise-robust radar HRR target recognition based on complex Gaussian statistical models," in *Proc. IEEE CIE Int. Conf. Radar*, Oct. 2016, pp. 1–4.
- [19] V. M. Patel, G. R. Easley, D. M. Healy, and R. Chellappa, "Compressed synthetic aperture radar," *IEEE J. Sel. Topics Signal Process.*, vol. 4, no. 2, pp. 244–254, Feb. 2010.
- [20] D. L. Donoho, A. Maleki, and A. Montanari, "Message-passing algorithms for compressed sensing," *Proc. Nat. Acad. Sci. USA*, vol. 106, no. 45, pp. 18914–18919, Nov. 2009.
- [21] P. Vincent, H. Larochelle, Y. Bengio, and P. A. Manzagol, "Extracting and composing robust features with denoising autoencoders," in *Proc. 28th Int. Conf. Mach. Learn.*, Jul. 2008, pp. 1096–1103.
- [22] J. Chen, L. Du, H. He, and Y. Guo, "Convolutional factor analysis model with application to radar automatic target recognition," *Pattern Recognit.*, vol. 87, pp. 140–156, Mar. 2019.
- [23] M. I. Skolnik, *Radar Handbook*, 3rd. ed. New York, NY, USA: McGraw-Hill, 2008.
- [24] Z. Zhang, H. Wang, F. Xu, and Y. Q. Jin, "Complex-valued convolutional neural network and its application in polarimetric SAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 7177–7188, Sep. 2017.
- [25] R. Hänsch and O. Hellwich, "Classification of polarimetric SAR data by complex valued neural networks," in *Proc. ISPRS Hannover Workshop High-Resolution Earth Imag. Geospatial Inf.*, 2009, vol. 38, pp. 4–7.
- [26] H. Yu and J. Yang, "A direct LDA algorithm for high-dimensional data with application to face recognition," *Pattern Recognit.*, vol. 34, no. 10, pp. 2067–2070, Oct. 2001.
- [27] K. Simonyan and A. Vedaldi, "Deep inside convolutional networks: Visualising image classification models and saliency maps," in *Proc. Int. Conf. Learn. Representations*, 2014, pp. 1–14.

- [28] A. Mordvintsev, C. Olah, and M. Tyka, "Inceptionism: Going deeper into neural networks," 2016. [Online] Available: <http://googleresearch.blogspot.com/2015/06/inceptionism-going-deeper-into-neural.html>
- [29] Q. Zhang, Y. N. Wu, and S. C. Zhu, "Interpretable convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 8827–8836.
- [30] S. Sabour, N. Frosst, and G. E. Hinton, "Dynamic routing between capsules," in *Proc. Adv. Conf. Neural Inf. Process. Sys.*, 2017, pp. 3859–3869.
- [31] Z. Zhang, Y. Liu, J. Liu, F. Wen, and C. Zhu, "AMP-Net: Denoising based deep unfolding for compressive image sensing," *IEEE Trans. Image Process.*, vol. 30, pp. 1487–1500, Dec. 2021.
- [32] Q. Kai, X. Mao, X. Shen, X. Wang, T. Li, and Y. Gu, "Time-varying graph signal reconstruction," *IEEE J. Sel. Topics Signal Process.*, vol. 1, no. 1, pp. 870–883, Jul. 2017.
- [33] M. Arnaudon, F. Barbaresco, and Y. Le, "Riemannian medians and means with applications to radar signal processing," *IEEE J. Sel. Topics Signal Process.*, vol. 7, no. 4, pp. 595–604, May 2013.
- [34] X. Desquesnes and A. Elmoataz, "Non-monotonic front propagation on weighted graphs with applications in image processing and high-dimensional data classification," *IEEE J. Sel. Topics Signal Process.*, vol. 11, no. 6, pp. 897–907, Jul. 2017.
- [35] C. Tison, N. Pourthie, and J. C. Souyris, "Target recognition in SAR images with support vector machines (SVM)," in *Proc. IEEE Int. Geosci. Remote Sensing, Symp.*, Barcelona, Spain, Jul. 2007, pp. 456–459.
- [36] W. Chen *et al.*, "Tensor RNN with Bayesian nonparametric mixture for radar HRRP modeling and target recognition," *IEEE Trans. Signal Process.*, vol. 69, pp. 1995–2009, Mar. 2021.
- [37] Z. Ying, C. Bo, Z. Hao, and Z. Wang, "Robust variational auto-encoder for radar HRRP target recognition," in *Proc. 7th Int. Conf., IScIDE*, Dalian, China, Sep. 2017, pp. 364–375.
- [38] H. Zhang, C. Bo, Z. Wang, and H. Liu, "Deep max-margin discriminant projection," *IEEE Trans. Cybern.*, vol. 49, no. 7, pp. 2454–2466, May 2019.
- [39] M. Pan, A. Liu, Y. Yu, P. Wang, and H. Zhu, "Radar HRRP target recognition model based on a stacked CNN-Bi-RNN with attention mechanism," *IEEE Trans. Geosci. Remo. Sens.*, vol. 60, Feb. 2022, Art. no. 5100814.
- [40] P. Kosir and R. Dewart, "Feature alignment techniques for pattern recognition," in *Proc. IEEE Nat. Aerosp. Electron. Conf.*, May 1994, pp. 128–132.
- [41] L. Du, H. Liu, Z. Bao, and J. Zhang, "A two-distribution compounded statistical model for radar HRRP target recognition," *IEEE Trans. Signal Process.*, vol. 54, no. 6, pp. 2226–2238, Jun. 2006.
- [42] J. Li and P. Stoica, "Efficient mixed-spectrum estimation with applications to target feature extraction," *IEEE Trans. Signal Process.*, vol. 44, no. 2, pp. 281–295, Feb. 1996.
- [43] Q. S. Wu, Y. M. Zhang, M. G. Amin, and B. Himed, "High-resolution passive SAR imaging exploiting structured Bayesian compressive sensing," *IEEE J. Sel. Topics Signal Process.*, vol. 9, no. 8, pp. 1484–1497, Sep. 2015.



Leiyao Liao received the B.Eng. degree in electrical engineering in 2017 from Xidian University, Xi'an, China, where she is currently working toward the Ph.D. degree in signal processing with the National Laboratory of Radar Signal Processing. Her research interests include radar automatic target recognition and pattern recognition.



Lan Du (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electronic engineering from Xidian University, Xi'an, China, in July 2001, March 2004, and June 2007, respectively. Her doctoral dissertation was granted National Excellent Doctoral Dissertation of China in 2009. From September 2007 to September 2009, she did research work as a Postdoctoral Research Associate with the Department of Electrical and Computer Engineering, Duke University, Durham, NC, USA. She is currently a Professor with the National Laboratory of Radar

Signal Processing, Xidian University. Her main research interests include statistical signal processing and machine learning with application to radar target recognition. Her research work is supported by NSFC for Excellent Young Scholars.



Jian Chen received the B.Eng. degree in communication engineering from the Qingdao University of Technology, Qingdao, China, in June 2014, and the Ph.D. degree in electronic engineering from Xidian University, Xi'an, China, in August 2020, respectively. His research interests include radar signal processing, radar automatic target recognition, and pattern recognition.